AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals

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DEDICATION

This research is humbly dedicated to mute individuals and persons with speech impairments, whose daily struggles in communication inspired the development of this study. May this work serve as a small contribution toward creating a more inclusive society where their voices are heard, understood, and valued through the help of technology.

We also dedicate this study to our families and loved ones for their unwavering support, patience, and encouragement throughout this academic journey. Their guidance and motivation provided us with the strength to overcome challenges and complete this research.

To our thesis adviser and panel members, we express our sincere gratitude for their valuable insights, constructive criticisms, and continuous guidance, which greatly contributed to the improvement and completion of this study.

Finally, this research is dedicated to our institution, the Eulogio “Amang” Rodriguez Institute of Science and Technology (EARIST), for providing us with the knowledge, resources, and opportunities that shaped us as future Information Technology professionals committed to innovation and social responsibility.

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Thank you all from the bottom of their hearts.

C. P. B

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ABSTRACT

Title:	AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals
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Degree:	Bachelor of Science in Information Technology
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This study presents the system that utilizes Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision technologies to recognize hand gestures and sign language through a standard webcam or smartphone camera. Using frameworks such as MediaPipe and TensorFlow.js, the application processes real-time video input, classifies gestures through a trained Convolutional Neural Network (CNN) model, and converts recognized signs into readable text and synthesized speech via Text-to-Speech (TTS) functionality. The platform is web-based, eliminating the need for specialized

hardware or software installation, making it accessible across desktop and Android mobile browsers. Additionally, the study follows a developmental-descriptive research design using an iterative and incremental development approach. The system would be evaluated based on usability, accuracy, and accessibility through testing with selected mute individuals and guardians. Accuracy rate computation, System Usability Scale (SUS), and thematic analysis would be used to assess system performance and user experience. This study aligns with Sustainable Development Goal (SDG) 10 Reduced Inequalities, which aims to promote social, economic, and political inclusion for all, particularly marginalized and vulnerable groups such as persons with disabilities. By developing an AI-powered web application for real-time sign and gesture interpretation, the study seeks to reduce communication barriers faced by mute individuals in everyday interactions. The system promotes inclusivity, accessibility, and equal participation in education, healthcare, public services, and community engagement. With affordable and widely accessible technologies, the project supports the advancement of digital equity and empowers mute individuals to communicate independently, thereby contributing to the broader objective of reducing inequalities within society.

Keywords: Artificial Intelligence, Gesture Recognition, Filipino Sign Language, Computer Vision, Web-Based Application, Assistive Technology, Real-Time Processing.

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Communication is a vital human function that enables individuals to interact, share ideas, and participate in daily life. For people who are mute or have severe speech impairments, expressing themselves can be a significant challenge especially in communities where alternative communication methods, such as sign language, are not widely understood. These communication barriers often result in social isolation, reduced access to essential services, and limited opportunities for education and employment.

While sign language serves as an effective communication tool, its benefits are often limited by the general public's unfamiliarity with it, as well as the lack of formal training among many mute individuals. Additionally, existing communication aids such as interpreters or assistive devices are frequently expensive, socially restrictive, or inaccessible to those living in resource-limited settings.

In response to these issues, this study proposes the design and development of a real-time web-based gesture and sign language interpretation system aimed at enhancing the communicative independence of mute individuals. The system is envisioned to utilize advancements in artificial intelligence and computer vision to interpret hand gestures through

standard digital devices. By doing so, it intends to offer a more inclusive, flexible, and practical solution for bridging communication gaps in modern society.

This research is grounded in the principles of accessibility, inclusivity, and digital equity. It seeks to empower mute individuals by providing them with a tool that promotes autonomy, social participation, and a higher quality of life. Furthermore, the study aligns with local and international efforts to improve assistive technologies and uphold the rights of Persons with Disabilities (PWDs), particularly within urban communities where targeted support for non-verbal individuals remains limited.

Ultimately, the project contributes to the broader goal of building a society where communication is a right enjoyed by all, regardless of physical ability or verbal capacity.

Background of Study

With rapid advancements in artificial intelligence, computer vision, and web technologies, it is now possible to develop real-time systems that can interpret hand gestures using only commonly available devices such as smartphones and webcams. Such innovations offer an accessible, cost-efficient alternative to traditional assistive tools, which are often expensive, unavailable, or require specialized hardware. Despite ongoing

efforts in sign language research, existing systems typically focus on limited datasets, offline platforms, or specialized equipment, leaving a gap for a widely accessible, browser-based solution tailored to local Filipino users.

This study responds to that gap by developing a real-time web-based gesture and sign language interpretation system designed specifically to support mute individuals. By leveraging AI-driven hand gesture recognition and text-to-speech technologies, the system aims to provide a practical, inclusive tool that enhances independent communication and bridges the gap between mute individuals and non-signers in everyday interactions.

With the growing capabilities of computer vision, machine learning, and web technologies, it is now possible to build systems that can interpret hand movements and gestures in real time. These technologies can be integrated into a web-based platform that is both accessible and cost-efficient, eliminating the need for expensive hardware or software installations. Through the use of standard webcams or smartphone cameras, such a system can detect, classify, and translate sign language or predefined gestures into text or spoken language.

Given the need for inclusive and accessible communication tools, this study aims to develop a real-time web-based gesture and sign language interpretation system specifically designed for individuals who are deaf or mute. The system will serve as an assistive communication tool, enabling

users to express themselves more freely and interact with others who do not understand sign language. By focusing on this geographic location, the study also considers the unique socio-economic and technological context of the local community.

In a world where verbal communication dominates everyday interactions, mute individuals face significant barriers to expressing their thoughts, needs, and emotion. Although sign language offers an effective medium for communication, its usefulness is limited by the general public's lack of understanding. This issue is particularly evident in busy urban areas, where quick and efficient communication is essential in public spaces, transportation, government offices, and health institutions.

With the advancement of web development, artificial intelligence (AI), and computer vision, there is now a growing opportunity to develop smart, real-time systems that can recognize hand gestures and sign language using just a webcam or camera. By integrating these technologies into a web-based platform, this project seeks to create an accessible, cost-effective, and user-friendly communication tool for mute individuals. The proposed project, a Real-Time Web-Based Gesture and Sign Language Interpretation System, seeks to address this communication gap using accessible and cost-effective technologies. By integrating computer vision and machine learning within a web application, the system can recognize and

interpret hand gestures and sign language in real-time, converting them into readable or spoken text with audio, but limited.

Purpose and Description

The primary purpose of this project is to design and develop a real-time web-based system that interprets hand gestures and sign language to aid mute individuals in effectively communicating with people who do not understand sign language. This system is intended to serve as an assistive communication tool that reduces the communication gap between mute individuals and the public.

By leveraging computer vision, machine learning, and web technologies, the project aims to create an accessible, user-friendly, and cost-effective platform that can recognize common signs or gestures and instantly convert them into text or speech output. The system is built to promote inclusivity, foster independence, and improve the social interaction and quality of life for individuals with speech impairments.

Project Description

The proposed system is a real-time, web-based gesture interpretation tool de- signed to support basic communication for mute individuals using hand gestures. The system captures live video through a webcam or smartphone camera and processes the visual input using AI-based computer vision techniques.

Because the system runs entirely inside a web browser on both mobile and desktop devices, it requires no installation or specialized hardware beyond a standard camera. This application is accessible, lightweight, and practical for everyday use. Overall, the project aims to provide a simple yet meaningful assistive communication feature for mute individuals while demonstrating the potential of AI-driven gesture recognition within real-time, browser-based environments.

Conceptual Framework

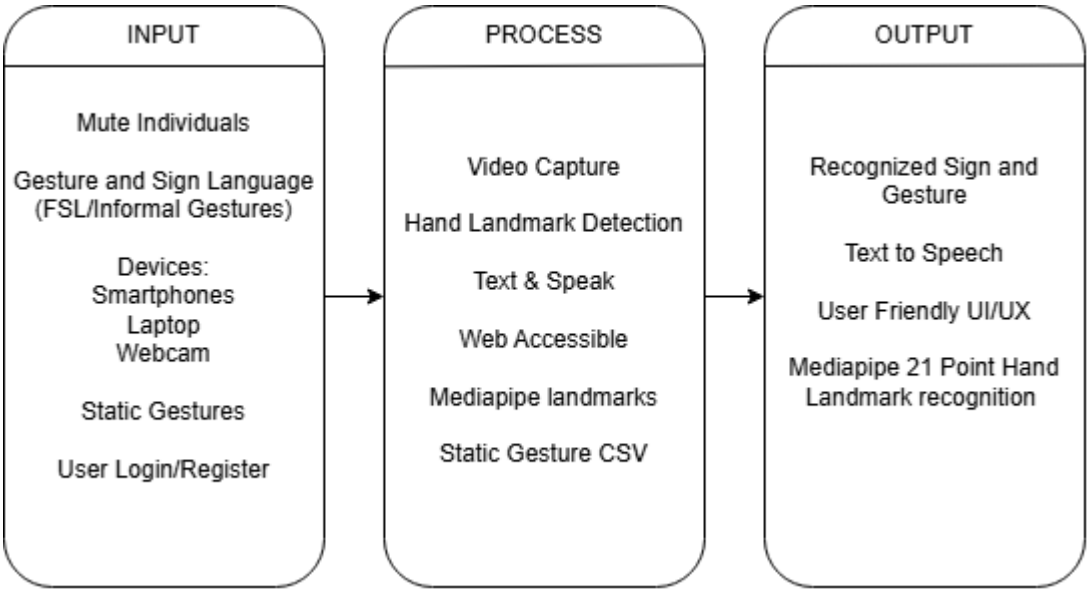

Figure 1.1 Conceptual Framework

Figure 1.1 This conceptual framework presents the fundamental structure of the proposed Real-Time Web-Based Gesture and Sign Language Interpretation System for Mute Individuals. The framework follows an Input, Process, Output (IPO) model, illustrating how user interactions, advanced

processing, and system outputs collectively enable accessible, inclusive communication.

Inputs include the active participation of mute individuals, gesture and sign language data (both formal, such as Filipino Sign Language, and informal or user- defined gestures), and the use of readily available digital devices, such as smartphones, laptops, and webcams. Additionally, the system incorporates secure user features, including login, registration, identity photo upload, and user privacy, and accommodates dependent users.

The Processing component represents the system's intelligent core: a real-time gesture and sign language interpretation module powered by Artificial Intelligence (AI) and computer vision. Through device cameras, user gestures are captured as video frames and processed via trained AI models.

Outputs are delivered in two accessible formats: Readable Text, which displays the interpreted message clearly on the user interface, and Synthesized Speech, which uses text-to-speech (TTS) technology to vocalize the recognized gestures. This dual-mode output empowers mute individuals to communicate effectively with non-signing people in real time.

Overall, this framework highlights the study's emphasis on developing an inclusive, responsive, and practical communication tool that directly addresses the communication barriers faced by individuals with speech

impairments. It reflects the system's core goal of facilitating real-time, two-way communication using familiar, low-cost technologies, making it a scalable and meaningful solution for local communities.

Objectives

The following specific objectives have been identified to guide the development, implementation, and evaluation of the system:

1. To develop a user-friendly web-based application capable of detecting and recognizing predefined static hand gestures using a standard webcam or smartphone camera.

2. To implement an AI-driven gesture recognition model that accurately identifies the supported static hand signs in real time using computer vision techniques.
    2.1. Static Hand Gesture Recognition Algorithm Using Hand Landmark Detection - The system applies to a vision-based static hand gesture recognition algorithm that detects hand landmarks from camera input to classify static sign letters.

3. To evaluate the system's usability, accuracy, and performance through testing sessions conducted with mute individuals or guardians within selected areas.
4. To acknowledge and document the system's detection limitations, particularly its inability to interpret dynamic or informal gestures, and to establish recommendations for future enhancement of gesture recognition capabilities.

5. To satisfy the quality of the system using ISO 25010 standards in terms of:
   - Suitability
   - Reliability
   - Usability
   - Security
   - Maintainability
   - Portability

Significance of Study

This study holds significance by demonstrating how AI-driven gesture recognition can provide simple yet meaningful communication aid for mute individuals despite limited gesture coverage. It introduces a foundational tool

that shows the feasibility of web-based assistive technology using common devices.

1. **Mute Individuals:** The system offers an accessible way for users to ex- press basic letters through hand gestures that are instantly converted into text or speech. While limited, it provides users with an initial tool to support simple communication needs.
2. **Families and Caregivers:** Caregivers benefit from an easier way to understand gestures performed by mute individuals, improving household communication, and assisting in basic daily interactions.
3. **Educators and Healthcare Workers:** The prototype demonstrates how basic gesture-recognition tools can be integrated into classrooms, therapy sessions, or clinics to support learning and communication for individuals with speech impairments.
4. **Researchers and Technologists:** The project serves as a proof of concept for browser-based gesture recognition. It provides a foundation for future expansions, such as recognizing more letters, dynamic gestures, or a complete sign-language vocabulary.

Overall, this study aims to bridge communication gaps, raise awareness of gesture language, and promote inclusiveness using technology-enhanced learning and interaction.

Scope and Delimitations of the Study

This study focuses on the design and development of a real-time, browser-based gesture interpretation prototype intended to assist mute individuals in basic communication. The scope of the study includes the following: The scope of study includes the following:

Real-Time Gesture Interpretation: Using a webcam or smartphone camera, the system processes live video input and provides immediate output in real time.

Web-Based Accessibility: The system operates entirely through a web browser, making it accessible on Android mobile browsers and desktop devices without installation or specialized hardware.

Dual Output Format: Recognized gestures are converted into readable on- screen text and optional synthesized speech through text-to-speech (TTS) technology.

Basic User Features: The system includes essential features such as user login, registration, and a simple user profile interface.

Prototype-Level Testing: Evaluation focuses on usability, accuracy, and ease of use through testing sessions with selected participants.

Software Prototype: The study does not involve the development of physical devices, mobile applications, sensors, or hardware components.

This capstone project is focused on software design and prototyping only and will not include the development of any physical devices, mobile applications, or advanced hardware-driven module.

This study is limited to the interpretation of predefined static hand gestures only. The system relies on Media Pipe Hands, which detects hand landmarks but does not include a complete American Sign Language (ASL) or Filipino Sign Language (FSL) recognition model. The system does not support some dynamic gestures, two-hand signs, sequential signing, or any motion-based components of natural sign language. Non-manual cues, including facial expressions, head movements, lip patterns, and eye gaze, are also excluded, even though they are essential to full sign language meaning. Gesture accuracy is influenced by environmental conditions such as lighting, shadows, camera quality, hand distance, and background distractions. Although these factors are acknowledged in the system evaluation, optimizing performance for difficult environments is not included in this study.

Furthermore, the project does not involve training large-scale ASL/FSL datasets or developing custom machine learning models. The system functions strictly within the limitations of browser-based AI tools and does not attempt to provide full sign language translation or natural language interpretation.

Definition of Terms

Computer Vision – A field of artificial intelligence that enables computers to interpret and process visual information from the world, such as images or video frames.

Gesture Recognition – A technology that interprets human hand or body movements through visual or sensor input, allowing computers to understand and respond to human gestures.

Real-Time Processing – The ability of the system to detect, interpret, and dis- play the output (text or speech) of a gesture or sign as it is being performed, with minimal delay.

Sign Language – A visual language that uses hand shapes, facial expressions, and body movements to convey meaning. This study may focus on commonly used gestures and elements of Filipino Sign Language (FSL).

Text-to-Speech (TTS) – A technology that converts written text into spoken words. This feature helps bridge communication between mute individuals and hearing individuals.

User Interface (UI) – The visual part of the web application through which users interact with the system. A good UI is intuitive, responsive, and easy to navigate.

Chapter 2

REVIEW OF RELATED WORKS

This chapter presents a review of related literature and studies relevant to the development of the AI-Powered Web Application for Real-Time Sign and Gesture Interpretation System. These works provide theoretical and technical foundations for gesture recognition, computer vision, machine learning, and assistive communication technologies. The review includes both local and foreign studies to highlight existing solutions, limitations, and research gaps that support the need for the proposed system.

Local Literature

A study conducted by Piner et al. (2024) from the University of San Jose-Recoletos developed a Filipino Sign Language (FSL) recognition system using Media Pipe and machine learning techniques. The system used hand landmark detection to classify gestures and improve communication between speech-impaired individuals and the public. The researchers emphasized that Media Pipe significantly enhances gesture recognition accuracy while maintaining real-time processing capability, making it suitable for assistive communication applications.

More recent studies, such as Villanueva et al. (2021) from Mapua University, explored webcam-based FSL gesture detection using CNNs. Although their system achieved accurate recognition of static signs, the researchers reported difficulty identifying dynamic gestures such as J and Z—gestures that require motion and multiple frames to interpret. This limitation parallels the capabilities of the present study, which also focuses on static gestures only due to constraints in dataset size, model complexity, and real-time processing requirements.

Overall, local literature consistently demonstrates that gesture recognition systems in the Philippines tend to support small, static gesture datasets, often limited to a subset of the alphabet. These studies show promising potential but highlight challenges in recognizing dynamic gestures, environmental variations, and full sign-language vocabulary—gaps which the current project also acknowledges while contributing a browser-based.

Recent research also explored hybrid deep learning models such as CNN-LSTM for gesture recognition. These models combine spatial and temporal feature analysis to improve gesture classification accuracy. Studies have shown that hybrid models achieve approximately 92% accuracy in recognizing sign language gestures, demonstrating their effectiveness for real-time gesture recognition systems.

Foreign Literature

Sarma et al. (2025) conducted a study using Media Pipe and machine learning algorithms to recognize American Sign Language alphabets. Their system utilized a dataset of over 87,000 gesture images and evaluated multiple classification models. The Random Forest model achieved an accuracy of 98.83%, confirming the effectiveness of landmark-based gesture recognition in real-time communication systems. Recent research continues to explore mobile-based and browser-friendly systems. A 2022 CNN-based Indian Sign Language study demonstrated successful static-sign detection but was limited in recognizing motion-dependent gestures, similar to the constraints of the current project. These studies highlight that even globally, systems that rely solely on 2D video streams struggle with dynamic signs, reinforcing the relevance of focusing on static gestures for prototype development. Another study by Saleh (2025) compared Convolutional Neural Networks (CNN) and Support Vector Machine (SVM) for sign language recognition using Media Pipe landmarks. The CNN model achieved 100% classification accuracy, outperforming SVM in gesture detection. The study concluded that deep learning models provide higher accuracy and better generalization for gesture recognition systems.

In addition, Uddin et al. (2025) developed a Norwegian Sign Language recognition system using Media Pipe and Long Short-Term Memory (LSTM) networks. The system successfully extracted gesture features and recognized numerical signs, demonstrating that combining landmark detection and sequence-based models improves gesture interpretation accuracy and real-time processing capability.

Local Studies

Several Filipino academic institutions have explored gesture recognition as a tool for accessibility. The University of the East (2020) created a MATLAB-based FSL recognition model that identified a small number of static gestures. Despite its simplicity, the study demonstrated that even low-cost systems can support basic communication tasks.

Mapua University's 2021 study developed a CNN-powered recognition model using webcams to interpret FSL hand signs. Although effective for static letters, the researchers reported accuracy issues when encountering gesture variations or lighting inconsistencies—common challenges also observed in the present project.

Another local study by Santos et al. (2020) developed a gesture-based communication tool using Android devices. The system worked with a limited set of static signs and emphasized user accessibility. However, it was not

web-based and required application installation, highlighting a gap in browser-based assistive technologies that the current study aims to address.

These local studies consistently reveal that Philippine-based systems often support only small, static gesture sets due to dataset limitations, hardware constraints, and resource availability, aligning closely with the design of the present prototype.

Foreign Studies

Foreign research typically involves more advanced approaches, combining deep learning, 3D sensors, or complex sequential models. Qi et al. (2021) introduced a deep-learning gesture recognition model optimized for contactless human–computer interaction, but its reliance on sequential motion tracking made it unsuitable for basic webcam-only setups.

A 2020 IEEE study using CNN-LSTM hybrid models improved dynamic gesture recognition accuracy but required more computational resources than real-time browser-based systems can provide. Stergiopoulou and Papamarkos (2009) applied neural-network shape-fitting techniques to static hand gestures, successfully recognizing letters with distinct shapes—like the approach taken in this project.

Lopez and Reyes (2022) explored wearable sensors for ASL recognition, achieving high accuracy but requiring specialized hardware such

as EMG and IMU devices. Chen et al. (2023) developed a sensor glove for real-time translation but noted its limited accessibility due to cost.

Globally, research shows that static gesture recognition remains the most feasible approach for lightweight, accessible prototypes, especially those relying solely on standard cameras and browser-based technologies.

Synthesis of Related Works

The reviewed literature and studies collectively demonstrate the significant role of computer vision, machine learning, and assistive technologies in improving communication for individuals with speech and hearing impairments. Both local and foreign research confirm that gesture recognition systems have evolved from basic image processing techniques to advanced deep learning models capable of interpreting hand gestures with high accuracy and real-time performance.

Local literature highlights the growing interest in developing Filipino Sign Language (FSL) recognition systems using accessible and low-cost technologies. Studies conducted by Piner et al. (2024), Villanueva et al. (2021), and other Philippine researchers show that webcam-based gesture recognition combined with machine learning and Media Pipe can successfully detect and classify static hand gestures. These studies emphasize the practicality of browser-based and camera-based systems in promoting

accessibility. However, local research consistently reports limitations such as small dataset sizes, difficulty recognizing dynamic gestures, and sensitivity to environmental factors like lighting and camera positioning. Additionally, many local systems are limited to static alphabet gestures and are not fully deployed as web-based real-time applications, highlighting a gap in accessible, scalable assistive communication tools.

Foreign literature and studies further support the effectiveness of machine learning and deep learning models in gesture recognition. Research conducted by Sarma et al. (2025), Saleh (2025), and Uddin et al. (2025) demonstrates that landmark-based detection using Media Pipe combined with classification models such as Random Forest, CNN, and LSTM can achieve high accuracy rates, often exceeding 95%. Foreign studies also explored hybrid models like CNN-LSTM, which improve gesture recognition by analyzing both spatial and temporal features. However, these systems often require large datasets, higher computational resources, or specialized hardware such as sensor gloves or wearable devices, making them less practical for lightweight, browser-based implementations.

Both local and foreign studies consistently reveal that static gesture recognition is more feasible and reliable for real-time systems using standard webcams, while dynamic gesture recognition remains challenging due to motion complexity, dataset requirements, and processing demands. Media Pipe has emerged as a widely used and effective tool for hand landmark

detection, providing accurate and efficient feature extraction for gesture classification. Despite advancements in deep learning and sensor-based systems, accessibility, cost, and deployment complexity remain key challenges in assistive communication technologies.

Based on these findings, the present study contributes by developing an AI-powered web-based gesture interpretation system that leverages computer vision and machine learning for real-time recognition using standard webcams. Unlike systems that require specialized hardware or offline processing, the proposed system focuses on accessibility, real-time performance, and browser-based deployment. By addressing limitations identified in previous studies—such as lack of web integration and accessibility constraints—the present research aims to provide a practical and scalable assistive communication solution while maintaining reliable recognition of static hand gestures.

Chapter 3

METHODOLOGY

Technical Background

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision (CV) have enabled the development of lightweight, accessible systems capable of interpreting hand gestures in real time using standard digital devices. These technologies provide new possibilities for supporting individuals with speech impairments by transforming visual hand inputs into meaningful outputs. However, many advanced systems require high-performance hardware or extensive datasets, making it difficult to apply in low-resource or browser-based environments. At the core of the system is Media Pipe Hands, a computer vision frame- work capable of extracting 21 three-dimensional hand landmarks from a live webcam feed. These landmarks provide geometric representation of the hand's shape and orientation. Instead of using heavy deep-learning models, the system uses a lightweight classifier (e.g., K-Nearest Neighbors or landmark-based distance matching) to compare detected hand shapes with predefined gesture samples.

This allows real-time classification directly in the browser using Tensor- Flow.js, without the need for GPU acceleration or external processing servers. Once a gesture is recognized, the system converts the output into

readable on-screen text and optionally into spoken audio using the browser's built-in Text-to- Speech (TTS) API. This dual output enhances accessibility, allowing mute individuals to communicate more effectively with non-signers. The system is fully web-based, meaning it operates on both mobile and desktop browsers without requiring installation or specialized hardware aside from the device's camera.

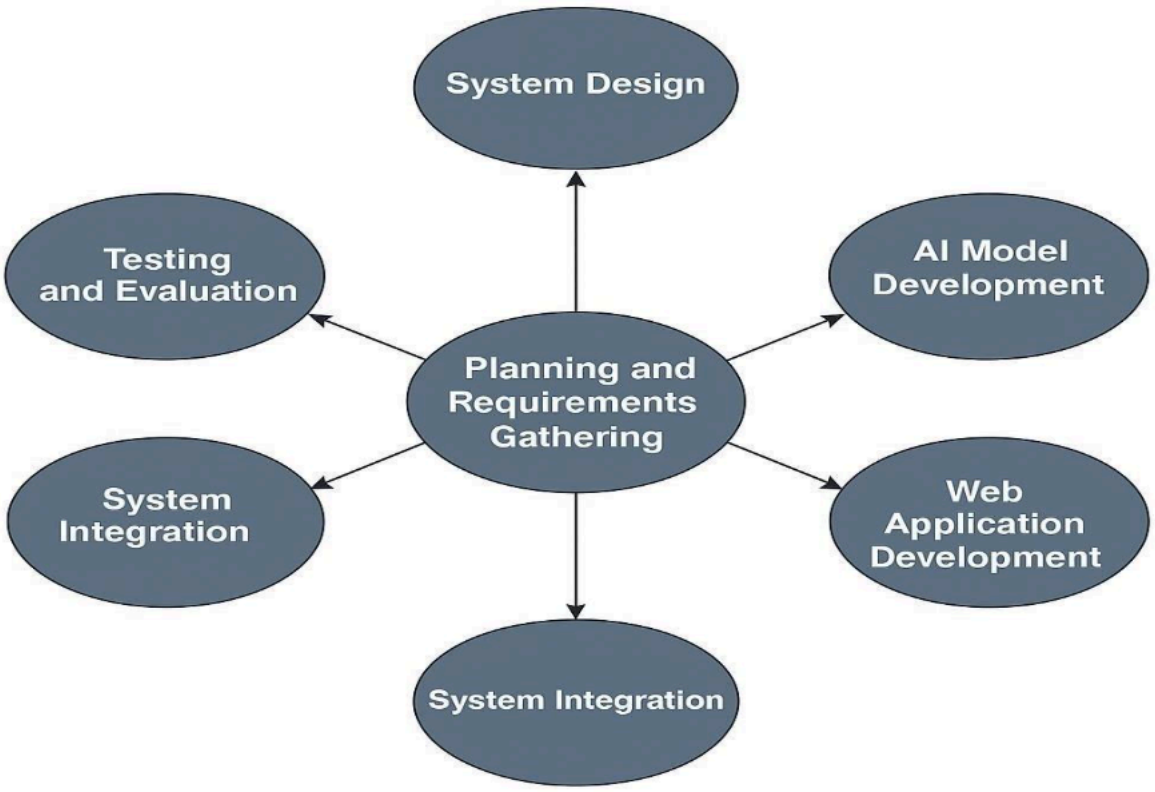

Figure 2. Development Methodology Diagram

Figure 2.2 The development methodology illustrated in the diagram presents a structured, sequential, and iterative process that guided the creation of Real-Time Web-Based Gesture Interpretation System. At the core of the diagram is Planning and Requirements Gathering, which establishes the foundation for the entire project by identifying user needs, functional and non-functional requirements, and the overall system scope. From this central phase, the process expands into six key development activities. System Design involves formulating the system architecture, interface layout, and technical workflow necessary to support stable real-time gesture recognition. AI Model Development focuses on preparing and training the static gesture recognition model using Computer Vision techniques appropriate for lightweight browser execution. In parallel, Web Application Development addresses the creation of both front-end and back-end components, ensuring platform accessibility across mobile and desktop devices. Once these components are built, System Integration consolidates all modules—gesture detection, classification, user interface, and text-to-speech output—into a unified functional prototype. This integrated system then undergoes Testing and Evaluation, where its performance, usability, and accuracy are assessed through structured user testing. Finally, the prototype proceeds to the System Deployment stage, where it is made accessible for real-world use and further refinement. This methodology flow ensures that the system is developed

through a logical, organized, and user-centered approach consistent with established software engineering practices.

Requirement Analysis

The Requirement Analysis identifies and defines the essential functionalities, constraints, and system expectations needed to develop the Real-Time Web- Based Static Gesture Interpretation System. It covers both functional requirements, which specify what the system must accomplish, and non-functional requirements, which outline the quality attributes necessary for effective operation.

1. **Functional Requirements:** The functional requirements describe the specific behaviors and feature the system must provide:
    1.1. **Gesture Detection** - The system must detect and interpret predefined static hand gestures. The system must capture hand gestures through a webcam or mobile phone camera.
    1.2. **Gesture Classification** - The system must use Media Pipe Hands to extract hand landmarks and classify gestures through a lightweight model. The system must provide real-time identification of gestures.
    1.3. **Text Output** - The system must display the recognized gesture as on-screen text immediately after detection.

1.4. **Speech Output (TTS)** - The system must convert recognized gestures into spoken audio using the browsers.

1.5. **User Authentication** - The system must allow users to register an account with basic information.

1.6. **Real-Time Video Preview** - The system must show a live feed of the user's hand to guide gesture positioning.

1.7. **Data Handling** - The system must not store video streams or sensitive gesture data.

1.8. **System Compatibility** - The system must run directly in a web browser without installation. The system must support both mobile and desktop environments.

2. **Non-Functional Requirements:** The non-functional requirements specify quality standards that the system must meet:

2.1. **Performance** - Gesture recognition must operate in real time with minimal delay.

2.2. **Usability** - The interface must be simple, clean, and easy to navigate.

2.3. **Reliability** - The system must consistently detect gestures within its supported set.

2.4. **Security** - All login and registration data must be encrypted or handled securely.

2.5. **Compatibility** - The system must function on major web browsers (Chrome, Edge, Firefox).

2.6. **Maintainability** - Code must be modular to allow for future expansion (e.g., adding more gestures).

Population, Sample Size, and Sampling

The target population of the study consists of mute individuals residing in Pasay City, particularly those who rely on basic hand gestures for daily communication. Since some mute individuals may require support in using digital tools, the population also includes guardians or family members who assist them during the evaluation of the system. This ensures that the system is tested within its intended user group and under real-life usage conditions.

Sample Size

- Use hand gestures as part of their daily communication;
- Willing to participate in system testing and provide feedback.

Sampling Technique

The study employs Purposive Sampling, a non-probability sampling method wherein participants are intentionally selected based on specific characteristics relevant to the study. Individuals were chosen because they:

Are mute or have speech impairments

Use hand gestures as part of their daily communication;

Willing to participate in system testing and provide feedback

This technique ensures that the participants are genuinely representative of the system's intended users. Guardians or family members may be included under the same sampling method if the mute individual requires assistance in navigating the system or understanding the evaluation procedures.

Research Instruments

The study utilized several research instruments to systematically gather both quantitative and qualitative data regarding the performance, usability, and overall effectiveness of the Real-Time Web-Based Static Gesture Interpretation System. These instruments ensure the reliability and validity of the findings.

Prototype System

The primary instrument used in the study is the gesture interpretation prototype itself, which recognizes static hand gestures and converts them into text and optional speech output.

Accuracy Testing Checklist

An accuracy checklist was used to record the number of correctly and in- correctly recognized gestures during the testing phase. Participants

performed each of the gestures multiple times, and the researcher documented the system's recognition outcome.

Usability Evaluation Questionnaire (System Usability Scale – SUS) A standardized System Usability Scale (SUS) questionnaire was administered to measure participants perceived usability of the system. Guardians may assist mute participants in completing the questionnaire if needed.

Observation Checklist

An observation checklist was used during testing sessions to record non-verbal reactions, user behavior, system responsiveness, ease of interaction, difficulties encountered, and environmental factors affecting performance.

Semi-Structured Interview Guide

1. A semi-structured interview guide was used to gather qualitative feedback from participants and/or guardians after using the system. The guide included open-ended questions focusing on user experience, perceived benefits, limitations, and suggestions for improvement. This instrument provided deeper insights into user. To integrate essential user features such as login, registration, and

simple profile management while maintaining accessibility on both mobile and desktop browsers.

1. **Accuracy Rate Computation**

   Accuracy results were obtained from the Accuracy Testing Checklist, where correct and incorrect recognitions of the static gestures were recorded. The accuracy rate for each gesture and the overall accuracy of the system is computed using the formula:

$$\text{Accuracy (\%)} = \frac{\text{Number of Correct Recognitions}}{\text{Total Number of Attempts}} \times 100$$

2. **Descriptive Statistics**

   To support interpretation and presentation of results: Frequency counts were used to summarize correct and incorrect gesture recognitions.

   Percentages were used for accuracy, user acceptance, and observed issues.

   Averages (mean values) were computed for SUS results when multiple participants were involved.

These statistics provide a clear and objective representation of the system's performance.

3. **Qualitative Data Analysis**

Qualitative data gathered from:

- Reviewing participant feedback and observation notes
- Identifying recurring ideas, challenges, or positive comments
- Grouping insights into meaningful categories (themes)
- Summarizing the themes to complement quantitative findings
- Thematic analysis provides deeper understanding of user experience, limitations, and potential improvements.

System Requirements

The System Requirements define the hardware, software, and environmental specifications necessary for the Real-Time Web-Based Static Gesture Interpretation System to function effectively. These requirements ensure that the system performs optimally during development, testing, and deployment.

1. Hardware Requirements
    1.1. Client-Side (User Device)
    1.2. Researcher/Developer Device
2. Software Requirements
    2.1. Web Browsers: Google Chrome (recommended), Microsoft Edge, Mozilla Firefox
    2.2. Back-End Software
    2.3. AI and Computer Vision Libraries
    2.4. Development Tools

Network Requirements

- Stable internet connection with at least 5 Mbps download/upload
- HTTPS connection for secure camera access
- Low latency environment to support real-time gesture processing

Environmental Requirements

The system performs best under the following conditions:

- Indoor lighting with sufficient brightness for clear hand visibility
- Neutral or uncluttered background to minimize visual noise
- Single-hand usage within camera frame
- Distance of 30–60 cm from camera for optimal landmark detection

- Minimal hand motion (static gesture requirement)

User Requirements

- Basic familiarity with smartphone or browser use
- Ability to perform clear static hand gestures

Chapter 4

RESULTS AND DISCUSSION

The system demonstrates successful implementation of AI-based gesture recognition within a browser environment. The integration of real-time gesture detection, text output, and speech synthesis confirms the feasibility of using computer vision and machine learning to support communication for mute individuals.

The results validate that the system can serve as an effective assistive communication tool, promoting accessibility, independence, and inclusivity. While improvements can be made in gesture coverage and accuracy optimization, the current prototype provides a strong foundation for future development of more advanced sign language interpretation systems.

Presentation of Objective 1:

To securely register new users into the system by collecting essential identity and login information. This module ensures that each user has a unique and authenticated ac- count, enabling personalized access to the Real-Time Web-Based Gesture and Sign Language Interpretation System.

To develop a user-friendly web-based application capable of detecting and recognizing predefined static hand gestures using a standard webcam or smartphone camera.

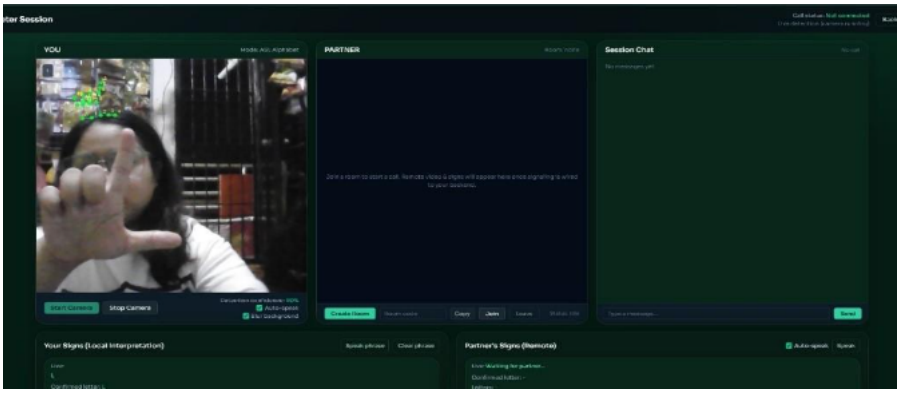

Figure 4.1 Interpreter Page

Figure 4.1. The interpreter page is where camera needs to be open and start the hand-posture. reshape it by the letter of any alphabet in static American sign language, camera would be detecting the landmark in the hand and it has confidence if the confidence is higher percentage, it will predict and confirmed the handshape letter, until its complete a word to speech the phrase using the text to speech output.

Presentation of Objective 2:

To implement an AI-driven gesture recognition model that accurately identifies the supported static hand signs in real time using computer vision techniques.

Static Hand Gesture Recognition Algorithm Using Hand Landmark Detection. - The system applies a vision-based static hand gesture recognition algorithm that detects hand landmarks from camera input to classify static sign letters.

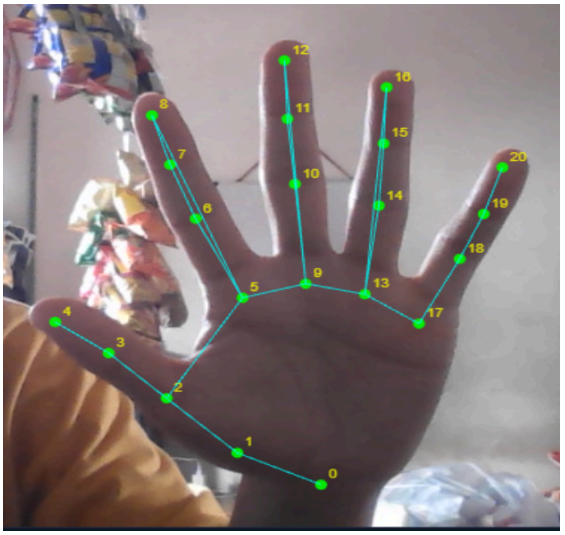

Figure 4.2 21 Points Landmarks

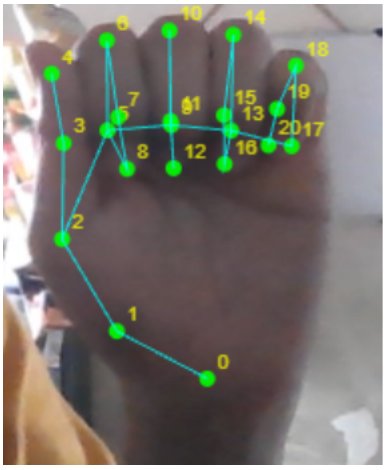

Figure 4.3 Hand Shape

Presentation of Objective 3:

To evaluate the system's usability, accuracy, and performance through testing sessions conducted with mute individuals or guardians within selected areas.

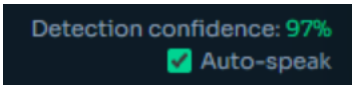

Figure 4.4 Accuracy Confidence Prediction

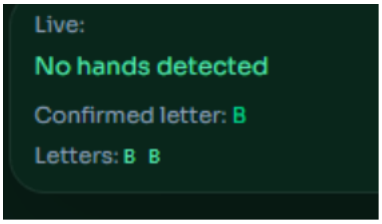

Figure 4.5 If no hand shown in the camera

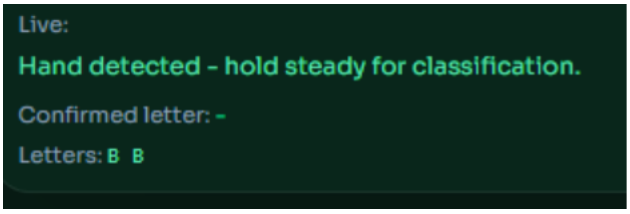

Figure 4.6 Hand shown in camera

Presentation of Objective 4:

To acknowledge and document the system's detection limitations, particularly its inability to interpret dynamic or informal

gestures, and to establish recommendations for future enhancement of gesture recognition capabilities.

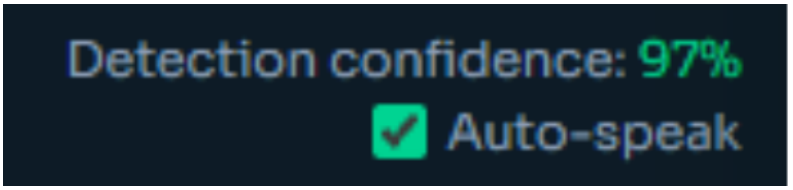

Figure 4.7 Detection Confidence Level

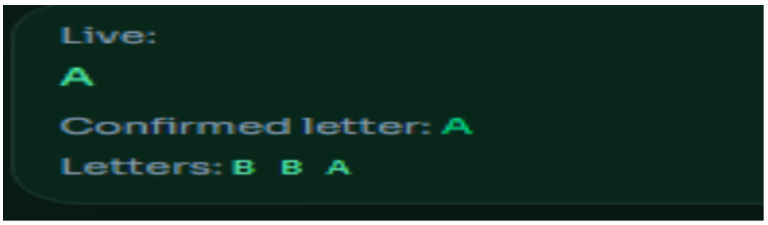

Figure 4.8 Confirmed Predicted Letter

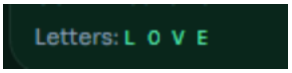

Figure 4.9 Confirmed Phrase

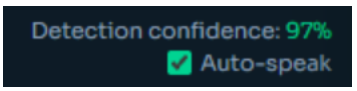

Figure 4.10 Text to Speech

Figure 4.11 Some of these features aren't accurate but can be detected as well within 1.7 to 2.5 seconds. In the future I recommend focusing on the accuracy of prediction, since the landmark 21 points need to be visible in the hand and if it can't be read by the prediction, it confused on what letter to detect.

Presentation of Objective 5:

Functional Suitability

Table 1

Assessment of Functional Suitability

Indicators	Weighted Mean	Verbal Interpretation	Rank
1. The features provided by the system are useful.	4.13	Agree	1
2. The system performs the tasks it is designed to do.	4.02	Agree	3
3. The system helps make tasks easier to finish.	4.11	Agree	2
Overall Weighted Mean	**4.13**	**Agree**	

As stated in Table 1, the assessment of functional suitability indicates that respondents generally agreed that the system effectively meets its intended purpose, as reflected by the overall weighted mean of 4.13 interpreted as Agree. Among the indicators, the statement that the features provided by the system are useful obtained the highest rank with a weighted mean of 4.13, suggesting that users recognize the value and relevance of the system's functions in supporting their tasks. This was followed by the perception that the system helps make tasks easier to finish with a weighted mean of 4.11, implying that the system contributes to efficiency and improved workflow. Meanwhile, the indicator stating that the system performs the tasks it is designed to do ranked third with a weighted mean of 4.02, which still reflects positive user confidence in system performance. The results imply that the system demonstrates strong functional suitability, as users perceive it to be practical, supportive, and aligned with their needs. Overall, it can be concluded that the system's functions are reliable and beneficial, indicating that it is capable of delivering its intended outcomes and supporting users in completing their tasks effectively.

Performance Efficiency

Table 2

Assessment of Performance Efficiency

Indicators	Weighted Mean	Verbal Interpretation	Rank

1. The system works smoothly without major delays.	4.00	Agree	2
2. The system meets my expectations.	4.07	Agree	1
3. The system loads pages and responds quickly.	3.89	Agree	3
Overall Weighted Mean	**3.99**	**Agree**	

As enumerated in Table 2, the assessment of performance efficiency shows that respondents agreed that the system operates efficiently, with an overall weighted mean of 3.99 interpreted as Agree. The indicator stating that the system meets my expectations ranked first with a weighted mean of 4.07, indicating that users generally perceive the system as capable of delivering satisfactory performance aligned with their needs. This was followed by the statement that the system works smoothly without major delays with a weighted mean of 4.00, suggesting that the system maintains stable operation during use. Meanwhile, the indicator that the system loads pages and responds quickly obtained the lowest rank with a weighted mean of 3.89, implying that while performance is acceptable, there is still room for improvement in speed and responsiveness. The findings imply that the system demonstrates adequate performance efficiency and is considered reliable by users, although enhancements in response time may further

strengthen user experience. Overall, it can be concluded that the system performs efficiently and meets user expectations, supporting consistent and satisfactory task completion.

Compatibility

Table 3

Assessment of Compatibility

Indicators	Weighted Mean	Verbal Interpretation	Rank
1. The system responds quickly when interpreting gestures.	3.82	Agree	2
2. The gesture recognition has little to no delay during use.	4.22	Strongly Agree	1
Overall Weighted Mean	**4.02**	**Agree**	

As indicated in Table3, the assessment of compatibility reveals that respondents generally agreed that the system performs well in terms of gesture-based interaction, as shown by the overall weighted mean of 4.02 interpreted as Agree. The indicator stating that the gesture recognition has little to no delay during use ranked first with a weighted mean of 4.22 interpreted as Strongly Agree, indicating that users experienced smooth and timely gesture recognition that supports uninterrupted interaction. The

statement that the system responds quickly when interpreting gestures ranked second with a weighted mean of 3.82 interpreted as Agree, suggesting that responsiveness remains satisfactory although not as strongly perceived as the minimal delay in gesture recognition. The results imply that the system is compatible with its gesture recognition feature and can deliver stable performance during actual use, which strengthens user confidence and usability in real-time tasks. Overall, it can be concluded that the system demonstrates good compatibility through efficient gesture recognition performance, contributing to a more seamless and user-friendly experience.

Usability

Table 4

Assessment of Usability

Indicators	Weighted Mean	Verbal Interpretation	Rank
1. The system is easy to learn.	4.02	Agree	5
2. The system is simple and easy to use.	4.11	Agree	1.5
3. The design of the system is clear and understandable.	4.11	Agree	1.5

4. The system has attributes that make it easy to operate and control	4.07	Agree	3
5. The system's design is visually clear and pleasant.	4.04	Agree	4
Overall Weighted Mean	**4.07**	**Agree**	

As presented in Table 4, the assessment of usability shows that respondents agreed that the system is user-friendly, with an overall weighted mean of 4.07 interpreted as Agree. The highest-rated indicators were that the system is simple and easy to use and that the design of the system is clear and understandable, both obtaining a weighted mean of 4.11 and sharing the top rank, indicating that users found the interface straightforward and the system functions easy to follow. The indicator that the system has attributes that make it easy to operate and control ranked third with a weighted mean of 4.07, suggesting that users feel capable of navigating and managing the system with minimal difficulty. The system's design being visually clear and pleasant ranked fourth with a weighted mean of 4.04, implying that the visual presentation supports comfort and clarity during use. Meanwhile, the system being easy to learn ranked fifth with a weighted mean of 4.02, which still reflects agreement but suggests that initial learning may require a short adjustment period for some users. The findings imply that the system

demonstrates strong usability through simplicity, clarity, and ease of control, which can promote higher acceptance and consistent use. Overall, it can be concluded that the system is usable and well-designed, enabling users to understand, operate, and interact with it effectively.

Reliability

Table 5

Assessment of Reliability

Indicators	Weighted Mean	Verbal Interpretation	Rank
1. The system performs reliably each time.	4.16	Agree	1
2. The system is always accessible when needed.	4.02	Agree	2
Overall Weighted Mean	**4.09**	**Agree**	

As report in Table 5, the assessment of reliability indicates that respondents agreed that the system is dependable, with an overall weighted mean of 4.09 interpreted as Agree. The indicator that the system performs reliably each time ranked first with a weighted mean of 4.16, suggesting that users consistently experience stable and dependable system performance during repeated use. The statement that the system is always accessible when needed ranked second with a weighted mean of 4.02, implying that availability is generally satisfactory, though it is slightly less emphasized than

consistency of performance. The results imply that the system can be trusted to function properly across sessions and is sufficiently accessible to support continuous use, which strengthens user confidence and reduces potential interruptions in task completion. Overall, it can be concluded that the system demonstrates good reliability through consistent performance and acceptable accessibility, supporting dependable and uninterrupted system utilization.

Security

Table 6

Assessment of Security

Indicators	Weighted Mean	Verbal Interpretation	Rank
1. I feel safe when using the system.	4.36	Strongly Agree	1
2. I believe my information is handled properly.	4.27	Strongly Agree	2
Overall Weighted Mean	**4.31**	**Strongly Agree**	

As listed in Table 6, the assessment of security shows that respondents strongly agreed that the system is secure, as reflected by the overall weighted mean of 4.31 interpreted as Strongly Agree. The highest-ranked indicator is that respondents feel safe when using the system with a weighted mean of 4.36, indicating a high level of user confidence and comfort during system use. This was followed by the belief that their

information is handled properly with a weighted mean of 4.27, suggesting that users trust the system's data handling practices and perceive that their privacy and information protection are prioritized. The findings imply that the system has established strong user trust in terms of safety and information management, which can encourage continued use and higher acceptance among users. Overall, it can be concluded that the system demonstrates a high level of security, as users strongly perceive it as safe to use and capable of handling information responsibly.

Maintainability

Table 7

Assessment of Maintainability

Indicators	Weighted Mean	Verbal Interpretation	Rank
1. The system seems organized and easy to improve in the future.	4.27	Strongly Agree	1
2. The system works well even after updates or changes.	4.22	Strongly Agree	2
Overall Weighted Mean	**4.24**	**Strongly Agree**	

As reports in Table 7, the assessment of maintainability indicates that respondents strongly agreed that the system is well structured and adaptable, with an overall weighted mean of 4.24 interpreted as Strongly

Agree. The indicator stating that the system seems organized and easy to improve in the future ranked first with a weighted mean of 4.27, suggesting that users perceive the system design as flexible and capable of supporting future enhancements. This was followed by the statement that the system works well even after updates or changes with a weighted mean of 4.22, implying that the system maintains stable performance despite modifications or improvements. The findings imply that the system demonstrates strong maintainability through its organized structure and resilience during updates, which supports long-term usability and sustainability. Overall, it can be concluded that the system is highly maintainable, as it allows continuous improvement while preserving reliable performance over time.

Chapter 5

Summary, Conclusion and Recommendations

Based on the findings, development process, and limitations identified in this study, the researchers present the following recommendations to further enhance the Real-Time Web-Based Gesture and Sign Language Interpretation System for Mute Individuals. These recommendations are intended to guide future researchers, developers, institutions, and stakeholders who may wish to improve, expand, or adopt the system.

1. System Enhancement and Feature Expansion

   Future development of the system may include the expansion of the gesture and sign language dataset to support a wider range of signs, including dynamic gestures, continuous sign sequences, and additional Filipino Sign Language (FSL) vocabularies. Incorporating facial expressions and body posture recognition may also improve the accuracy and expressiveness of sign interpretation, as non-manual signals are an important component of sign language communication.

   Additionally, the integration of context-aware recognition and sentence-level translation is recommended to move beyond isolated gesture recognition toward more natural conversational interaction.

2. Model Improvement and Accuracy Optimization

   To improve recognition accuracy, future researchers are encouraged to utilize larger and more diverse datasets collected from users of different ages, hand sizes, and skin tones, as well as under varying lighting and background conditions. Advanced deep learning architectures such as CNN-LSTM hybrid models or transformer-based vision models may be explored to enhance temporal gesture recognition and reduce misclassification.

   Regular retraining and fine-tuning of the AI model using real-world user data (with proper consent and anonymization) is also recommended to continuously improve system performance.

3. Platform Scalability and Deployment

   Although the current system is web-based, future versions may benefit from the development of a dedicated mobile application for Android and iOS platforms to improve performance, offline functionality, and device-level optimization. Cloud-based deployment with scalable server infrastructure may also be

considered to support a larger number of concurrent users. Integration with public service systems such as hospitals, government offices, and educational institutions may further enhance the system's real-world applicability and social impact.

4. Accessibility and User Experience Improvements

   Future enhancements should focus on improving accessibility features by incorporating customizable user interface elements such as adjustable text size, contrast settings, simplified layouts, and multilingual support. Including interactive tutorials, visual gesture guides, and onboarding walkthroughs may help first-time users better understand how to use the system effectively. User feedback mechanisms should also be expanded to allow continuous usability evaluation and iterative refinement of the interface.

5. Security, Privacy, and Ethical Considerations

   As the system involves sensitive user data, future implementations should strengthen security measures by integrating multi-factor authentication, encrypted data storage, and stricter access controls. Compliance with data privacy regulations, such as the Philippine Data Privacy Act of 2012, should remain a priority.

Clear consent mechanisms and transparent data usage policies are recommended, particularly if the system is deployed in public or institutional environments.

6. Wider Testing and Evaluation

   The current study utilized a limited sample size due to time and resource constraints. Future studies are encouraged to conduct large-scale testing involving participants from different locations and backgrounds to validate system performance and generalizability. Long-term evaluation studies may also be conducted to assess the system's impact on communication effectiveness, social inclusion, and quality of life for mute individuals.

7. Academic and Research Extensions

   Future researchers may build upon this study by comparing different AI models, evaluating cross-cultural sign language adaptation, or integrating the system with other assistive technologies such as speech recognition or wearable devices. The system may also serve as a foundation for further research in human-computer interaction, accessibility engineering, and inclusive technology design.

Summary

The system was designed to assist mute individuals in communicating with others by recognizing hand gestures through a webcam and translating them into readable text in real time. The system utilized Media Pipe for hand landmark detection and a machine learning classification model to identify specific gestures.

Overall, the proposed recommendations aim to guide the continuous improvement and sustainability of the Web-Based Gesture and Sign Language Interpretation System. By addressing technical, usability, accessibility, and ethical considerations, future work can further strengthen the system's role as an inclusive communication tool that empowers mute individuals and promotes equal access to communication in modern society.

The web-based deployment also improved accessibility, making the system easier to use for a wider range of users. Despite the limitations, the study demonstrated that artificial intelligence and computer vision can be effectively applied to develop accessible assistive communication tools. The system provides a practical solution to improve communication between mute individuals and the public and contributes to the advancement of inclusive technology.

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Appendices

Appendix A

Adviser Acceptance Form

REPUBLIC OF THE PHILIPPINES
EULOGIO "AMANG" RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
NAGTAHAN, SAMPALOC, MANILA

COLLEGE OF COMPUTING STUDIES

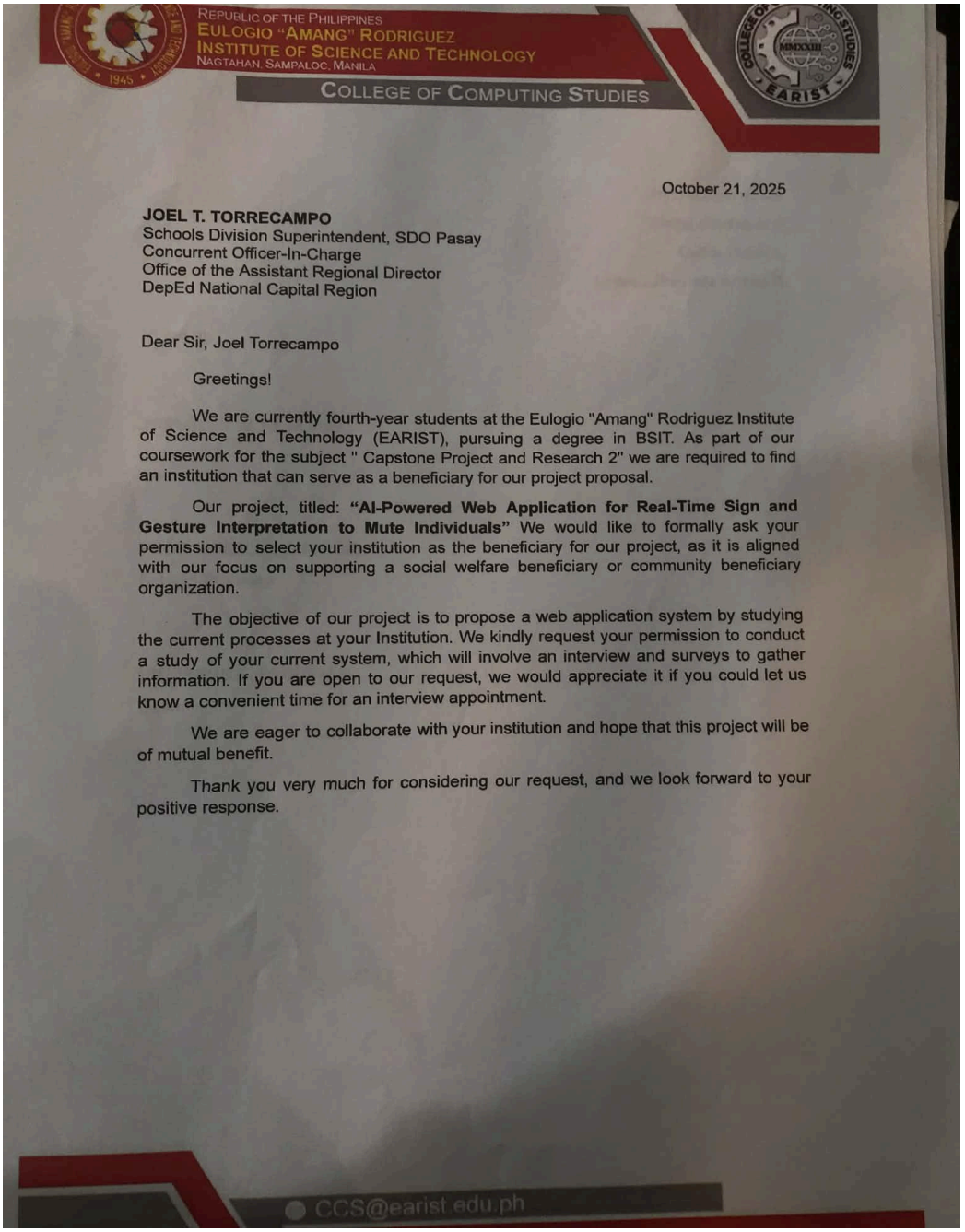
ACCEPTANCE OF THESIS ADVISER

NAME OF PROPONENTS:	Blaza, Christian P. Callo, Haley L. Lopez, Jheymie Shane A.
DEGREE PROGRAM:	BS INFOTECH
THESIS TITLE:	AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals
This is to certify that I accept to be the thesis adviser of the above-named students, and has validated their title as describe in this form.	
NAME AND SIGNATURE OF THESIS ADVISER:	Joevelyn W. Fajardo
RECOMMENDING APPROVAL: PROF. HAZEL F. ANUNCIO Subject Professor	APPROVED BY: DR. JESUS S. PAGUIGAN Dean

ccs@earist.edu.ph

Appendix B

Informed Consent to Beneficiary



REPUBLIC OF THE PHILIPPINES
EULOGIO "AMANG" RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
NAGTAHAN, SAMPALOC, MANILA

COLLEGE OF COMPUTING STUDIES

October 21, 2025

JOEL T. TORRECAMPO
Schools Division Superintendent, SDO Pasay
Concurrent Officer-In-Charge
Office of the Assistant Regional Director
DepEd National Capital Region

Dear Sir, Joel Torrecampo

Greetings!

We are currently fourth-year students at the Eulogio "Amang" Rodriguez Institute of Science and Technology (EARIST), pursuing a degree in BSIT. As part of our coursework for the subject " Capstone Project and Research 2" we are required to find an institution that can serve as a beneficiary for our project proposal.

Our project, titled: **"AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals"** We would like to formally ask your permission to select your institution as the beneficiary for our project, as it is aligned with our focus on supporting a social welfare beneficiary or community beneficiary organization.

The objective of our project is to propose a web application system by studying the current processes at your Institution. We kindly request your permission to conduct a study of your current system, which will involve an interview and surveys to gather information. If you are open to our request, we would appreciate it if you could let us know a convenient time for an interview appointment.

We are eager to collaborate with your institution and hope that this project will be of mutual benefit.

Thank you very much for considering our request, and we look forward to your positive response.

CCS@earist.edu.ph

Appendix C

Adviser Consultation Form

REPUBLIC OF THE PHILIPPINES
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INSTITUTE OF SCIENCE AND TECHNOLOGY
NAGTAHAN, SAMPALOC, MANILA

COLLEGE OF COMPUTING STUDIES

ADVISER CONSULTATION FORM

Name of the Researcher	Last Name First Name Middle Name 1. Blaza, Christian P. 2. Callo, Haley L. 3. Lopez, Jheymie Shane A.
Program	Bachelor of Science In Information Technology
Thesis Title	AI-Powered Web Based Application for Real-time Sign and Gesture Interpretation to Mute Individuals
Name of Adviser	Prof. Joevelyn W. Fajardo

Date	Consultation Concern	Agreement	Date of Next Consultation	Adviser Signature
5/21/25	First title	Revised Research Title		
5/28/25	second title	Finalize the Research Proposal		
6/11/25	Chapter 1-3	1st checking of content		

CCS@earist.edu.ph

Appendix D

Letter for Panelists

REPUBLIC OF THE PHILIPPINES
EULOGIO "AMANG" RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
NAGTAHAN, SAMPALOC, MANILA
COLLEGE OF COMPUTING STUDIES

December 1, 2025

PROF. EDMUND ALMAZAN

Sir,

I would highly appreciate it if you could be a member of the final defense panel that will evaluate the research work:

Title: **"AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals"**

PROPONENTS:	**Blaza, Christian P.**
	Callo, Haley L.
	Lopez, Jhaymie Shane A.
DATE:	**DECEMBER 5, 2025**
TIME:	**3:00 PM – 5:00PM**

Which is submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology.

I'm sending you a copy of the manuscript in this regard. Please notify the undersigned at least two (2) days before the scheduled oral presentation. If you find any discrepancy would require a postponement of their oral presentation.

Respectfully yours,
Ms. Hazel F. Anuncio
Thesis professor

CONFORME: 1/28/26
Prof. Edmund Almazan

CCS@earist.edu.ph

REPUBLIC OF THE PHILIPPINES
EULOGIO "AMANG" RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
NAGTAHAN, SAMPALOC, MANILA
COLLEGE OF COMPUTING STUDIES

December 1, 2025

PROF. ERNANIE CARLOS

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Title: **"AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals "**

PROPONENTS:	**Blaza, Christian P.**
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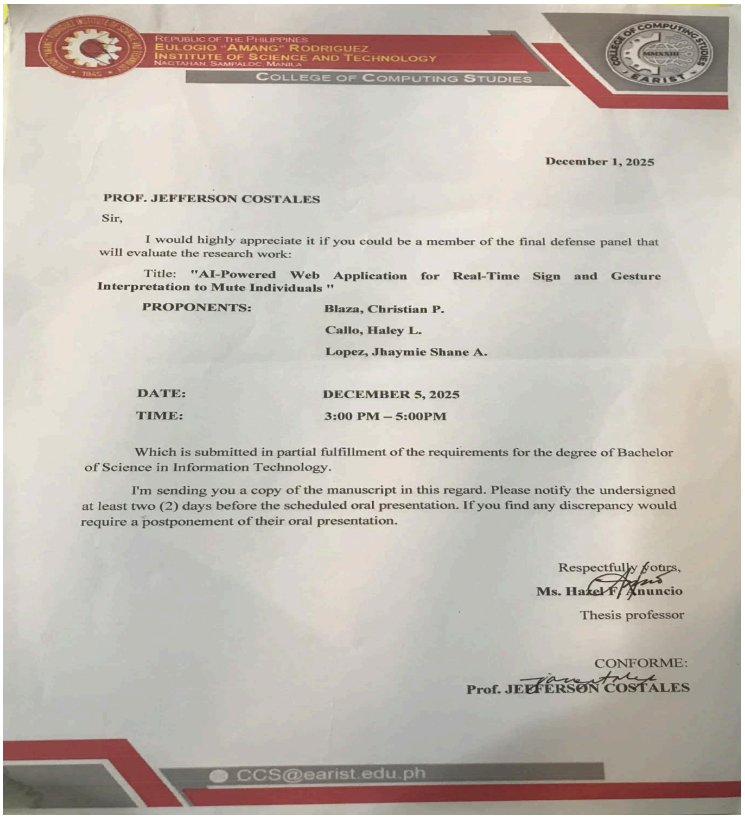
Respectfully yours,
Ms. Hazel F. Anuncio
Thesis professor

CONFORME:
Prof. ERNANIE CARLOS



REPUBLIC OF THE PHILIPPINES
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NAGTAHAN, SAMPALOC, MANILA
COLLEGE OF COMPUTING STUDIES

December 1, 2025

PROF. JEFFERSON COSTALES

Sir,

I would highly appreciate it if you could be a member of the final defense panel that will evaluate the research work:

Title: **"AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals "**

PROPONENTS: **Blaza, Christian P.**
Callo, Haley L.
Lopez, Jhaymie Shane A.

DATE: **DECEMBER 5, 2025**
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Respectfully yours,
Ms. Hazel T. Anuncio
Thesis professor

CONFORME:
Prof. JEFFERSON COSTALES

CCS@earist.edu.ph

Appendix E

Letter for Adviser

Appendix F

Recommendation

REPUBLIC OF THE PHILIPPINES
EULOGIO "AMANG" RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
NAGTAHAN, SAMPALOC, MANILA

COLLEGE OF COMPUTING STUDIES

Date: _____

☐ CS Thesis ☐ IT Capstone Project

TITLE: AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals

Student No.	Name of Student	Course/Yr/Sec
224-09261M	Blaza, Christian P.	BSIT 4-C
224-08832M	Callo, Haley L.	BSIT 4-C
224-08862M	Lopez, Jhaymie Shane A.	BSIT 4-C

IMPLEMENTATION OF THE RECOMMENDED REVISIONS

Nature of Revision	Previous	Revision/s Made	Page Reflected
	Make letter bigger	Make letter bigger overall feature	
	Separate quiz from accounts	Separated user analytics from user management in admin	
	Video/quiz uploadable	Video/quiz uploadable now	
		UI changes less dark	
		User video record itself for helping the system get sign accurate language in future, 1 handed or 2 handed	
		Revised feedback no more credentials needed since it's already logged in	

C Compiled NC Not Compiled	Examination Committee	Signature	Date
	Prof. Costales Jefferson		
C	Prof. Carlos Ernanie		2/2/26
C	Prof. Almazan Edmund		1/28/26

REVIEWED BY: : Prof. Costales Jefferson, Prof. Carlos Ernanie, Prof. Almazan Edmund

PROGRAM CHAIR : Prof. Hazel F. Annuncio DATE: Jan 27, 2026

DEAN : Dr. Jesus S. Paguigan DATE: Jan 27, 2026

Appendix G

Survey Questionnaire

ISO/IEC 25010:2011 - Systems and software engineering -- Systems and s

Functional suitability	SA	A	N	D	SD	Sub-Characteristics	Weighted Mean
	5	4	3	2	1		
Q1	8	6				Functional completeness	4.57
Q2	7	6	1			Functional correctness	4.43
Q3	9	4	1			Functional appropriateness	4.57
Performance efficiency	**SA**	**A**	**N**	**D**	**SD**	**Sub-Characteristics**	**Weighted Mean**
	5	4	3	2	1		
Q4	7	7				Time behaviour	4.50
Q5	9	5				Resource utilization	4.64
Q6	4	10				Capacity	4.29
Compatibility	**SA**	**A**	**N**	**D**	**SD**	**Sub-Characteristics**	**Weighted Mean**
	5	4	3	2	1		
Q7	6	7	1			Co-existence	4.36
Q8	8	5	1			Interoperability	4.50
Usability	**SA**	**A**	**N**	**D**	**SD**	**Sub-Characteristics**	**Weighted Mean**
	5	4	3	2	1		
Q9	4	7	3			Appropriateness recognizability	4.07
Q10	8	5	1			Learnability	4.50
Q11	7	7				Operability	4.50
Q12	8	5	1			User error protection	4.50
Q13	9	4	1			User interface aesthetics	4.57
Q14	6	7	1			Accessibility	4.36
Reliability	**SA**	**A**	**N**	**D**	**SD**	**Sub-Characteristics**	**Weighted Mean**
	5	4	3	2	1		
Q15	6	6	2			Availability	4.29
Q16	9	4	1			Fault Tolerance	4.57
Q17	9	4	1			Recoverability	4.57
Security	**SA**	**A**	**N**	**D**	**SD**	**Sub-Characteristics**	**Weighted Mean**
	5	4	3	2	1		
Q18	4	9	1			Confidentiality	4.21
Q19	7	6	1			Integrity	4.43
Q20	6	5	3			Non-repudiation	4.21
Q21	7	6	1			Accountability	4.43
Q22	5	9				Authenticity	4.36
Maintainability	**SA**	**A**	**N**	**D**	**SD**	**Sub-Characteristics**	**Weighted Mean**
	5	4	3	2	1		
Q23	8	6	1	3		Modularity	4.06
Q24	8	8	1	1		Reusability	4.28

Appendix H

Certificate of Statistician

Republic of the Philippines
Eulogio "Amang" Rodriguez
Institute of Science and Technology
Nagtahan, Sampaloc, Manila
College of Arts and Sciences

Certification

This is to certify that the following researchers:

Christian P. Blaza
Jheymie Shane A. Lopez
Haley L. Callo

from the Bachelor of Science in Information Technology program, consulted with and received advisement on the computation and interpretation of statistical tools and other relevant statistical procedures for their research paper entitled

"AI-Powered Web Application for Real-Time Sign and Gesture Interpretation to Mute Individuals"

a research at Eulogio "Amang" Rodriguez Institute of Science and Technology-Manila.

The consultation and advising pertain to the proper application and interpretation of statistical tools and/or treatments used in their research.

Given this 13th day of February 2026.

VISIONARY RESEARCH ASSOCIATION INC.
CERTIFICATE OF ACHIEVEMENT
Larex B. Tagalog
CERTIFIED STATISTICS SPECIALIST (CSS)™

LAREX B. TAGALOG, M.Sc, LPT, CSS™
Consultant (Statistical Treatment)
lbtagalog@earist.edu.ph
09162092169
Faculty Member of the College of Arts and Sciences

Appendix I

Certificate of Grammarian

Appendix J

Certificate of Plagiarism Scan

Appendix K

Short Report of Plagiarism Scan

Appendix L

User Acceptance Test

Appendix M

User Manual

Appendix N

Curriculum Vitae